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Participatory potentials in energy transition

A municipal communication ecology

Horsbøl, Anders; Pedersen, Anne Grethe Julius

Published in:
Energy Research & Social Science

DOI (link to publication from Publisher):
10.1016/j.erss.2026.104791

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Publication date:
2026

Document Version
Publisher's PDF, also known as Version of record

Link to publication from Aalborg University

Citation for published version (APA):
Horsbøl, A., & Pedersen, A. G. J. (2026). Participatory potentials in energy transition: A municipal communication ecology. *Energy Research & Social Science*, *137*, Article 104791. https://doi.org/10.1016/j.erss.2026.104791

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Energy Research & Social Science 137 (2026) 104791

Contents lists available at ScienceDirect

Energy Research & Social Science

journal homepage: www.elsevier.com/locate/erss

Original research article

Participatory potentials in energy transition: A municipal communication ecology

Anders Horsbøl *, Anne Grethe J. Pedersen

Department of Culture and Communication, Aalborg University, Denmark

ARTICLE INFO

Keywords:
Citizen participation
Communication ecology
Energy transition
Renewable energy
Discourse

ABSTRACT

In many countries, plans for renewable energy plants have sparked controversies and strong local opposition. In response, municipalities are launching new initiatives that emphasise stronger engagement of local communities. This study explores such a novel approach: the so-called Thy Model as developed by the Municipality of Thisted in northwestern Denmark. The Thy Model stands out for prioritising early community involvement and local support as essential prerequisites for planning renewable energy facilities such as wind turbines and large-scale photovoltaic plants. Building on a Communication Ecology framework, we view the Thy Model as an emerging communication ecology and analyse technological, social, and discursive layers of interaction at municipal and project site levels, where renewable energy projects are debated. Findings from the study, conducted between 2024 and 2025, show that the Thy Model introduces novel actor constellations, interaction formats, and communicative spaces. We argue that the Thy Model holds significant participatory potentials by enabling new networks of communication between local citizens and project developers that support local capacity building and address existing power imbalances. However, the study also reveals dilemmas of community representation and complex trade-offs between inclusion and exclusion in participatory processes.

1. Introduction[1]

Mitigating climate change critically depends on replacing fossil fuels with renewable energy (RE) sources such as wind and solar power. However, in many countries, plans for wind turbines and photovoltaic plants have been delayed or abandoned due to controversy and local resistance. Existing research identifies several key drivers for this resistance, including insufficient involvement of local communities, limited opportunities for community ownership, and perceived inequities in the distribution of benefits and burdens [1,2].

Similarly, in Denmark, plans for new RE plants have sparked controversy and strong local opposition, despite widespread public recognition of climate change as a pressing issue [3]. Opposition often centres on concerns about nuisance issues affecting neighbouring residents such as noise and shadow flicker. It also includes broader impacts on landscape, biodiversity, property values, the long-term viability of rural communities, settlement patterns, schools, childcare facilities, and local associations [4–6]. Moreover, critics question the democratic quality of decision-making processes, with many citizens feeling they have been "run over" by decision-makers [7]. Finally, some citizens argue that their community has already "done its part" and reached a saturation point for RE development.

In response to such controversy and resistance, several Danish municipalities have revised their approaches to RE planning and citizen participation. Between 2019 and 2021, all 98 Danish municipalities launched climate action plans. According to reports on Danish municipalities' climate initiatives, 88 of 98 Danish municipalities have prioritised climate initiatives, 55 municipalities have initiated 68 prioritised measures within RE, and 80% of the municipalities have introduced new initiatives to involve citizens in climate efforts [8,9]. The new strategies emphasise early involvement and stronger engagement of local communities in the energy transition, and many of these efforts go beyond conventional public consultation meetings, introducing new forms of organisation that aim to give local communities a stronger voice. The underlying rationale is that fundamental changes affecting everyday life require public engagement and local support to succeed and gain social acceptance.

A notable example is Thisted Municipality, located in the

* Corresponding author: Department of Communication and Psychology, Aalborg University, Rendsburggade 14, 9000, Aalborg, Denmark.
E-mail address: horsboel@ikk.aau.dk (A. Horsbøl).

[1] We would like to thank the editors and reviewers for their valuable feedback. We also gratefully acknowledge our colleagues and collaboration partners in the DEEP project, funded by Innovation Fund Denmark.

https://doi.org/10.1016/j.erss.2026.104791
Received 25 October 2025; Received in revised form 18 May 2026; Accepted 31 May 2026
Available online 4 June 2026
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northwestern region of Denmark known as Thy. In 2023, the municipal council adopted what is now referred to as the Thy Model, signalling a paradigmatic shift in citizen participation from reactive public consultations dominated by negative feedback to a proactive model centred on dialogue, early engagement, new community organisations, and local support and co-ownership as prerequisites for project approval.

This study examines the participatory potentials of the Thy Model by analysing it as an emerging communication ecology, i.e. a space of interconnected actors with specific social, technological, and discursive characteristics. Adopting a communication ecology perspective shifts the focus from isolated communication events, such as citizen meetings, to a broader constellation of actors, communication formats and discourses, distributed across time and space. The framework not only provides a strong foundation for identifying what is distinctive about the Thy Model and for assessing its participatory potentials and limitations but also provides municipalities and other social actors with insights into how communication infrastructures can foster or constrain citizen participation in RE planning. Moreover, the framework invites comparative analysis at regional, national, and international levels.

Specifically, we address the following research questions:

RQ 1: How are the key dimensions – technological, social, and discursive – of the Thy Model's communication ecology shaped?
RQ 2: What are the potentials and limitations for citizen participation within this communication ecology?

We examine the communication ecology at two levels; the broader municipal level and a specific project site where a proposed RE plant is under debate.

The remainder of the article is structured into five sections. We begin in Section 2 by reviewing existing research on citizen participation and opposition in relation to RE, situating the present study in the scholarly field, and presenting the communication ecology framework. In Section 3, we then provide an overview of the role of municipalities in Denmark's climate politics. In Section 4, we describe the research design, case, and data sources. This is followed by the analysis in Section 5, and the article concludes with a discussion of findings and their implications in Section 6.

2. Citizen participation and opposition in the renewable energy transition

In the literature on citizen involvement in the green energy transition, opposition to RE plants is well documented. Meta-studies highlight that citizens place strong emphasis on fair, transparent decision-making processes and an equitable distribution of benefits and burdens [1,10]. In a large study on social opposition to energy infrastructure, covering 130 cases across 3 continents, Sovacool et al. [11] observe that RE and low-carbon infrastructures are frequently protested. In a Scandinavian context, controversies over RE plants, particularly the siting of wind turbines, have recently been examined from various perspectives, including discourse, framing and narrative analyses [5,6,12,13].

Although the notion that opposition stems from NIMBYism (Not-In-My-Back-Yard) remains widespread in public discourse, this explanation has been widely critiqued in academic literature for oversimplifying local resistance as self-serving or hypocritical [14,15]. Alternative ways of viewing citizen resistance involve concerns for justice and democracy. Energy justice [16] is commonly divided into distributive, procedural, and recognition justice [17]. Distributive justice refers to the distribution of benefits and disadvantages, whether economic, environmental or social, including opportunities for local ownership and the generation of local value [18]. Procedural justice concerns the process of decision-making, including the procedural organisation and forms of involvement. Finally, recognition justice is about who is included as legitimate participants and how different identities are acknowledged [17]. Theoretically, the justice perspective replaces the instrumental view of citizen protests as an obstacle with a normative framework that opens for discussions of legitimate outcomes and processes. Empirically, it is supported by studies in different countries showing connections between perceived justice and social acceptance of RE projects [2,18,19,20].

The justice perspective has parallels to a democratic perspective, in particular a deliberative version that emphasises dialogue and the weighing of arguments [21]. However, more field specific notions such as ‘energy democracy’ [22] and ‘landscape democracy’ [23] do not provide a straightforward recipe for a democratic RE transition but rather a language for conceptualizing complexity and for navigating among different normative concerns. For instance, Arler et al. [23] points to three sets of values essential for a landscape democracy: personal freedom and self-determination, co-determination and participation, and objectivity and impartiality. While these values are interlinked, they do not necessarily point in the same direction, as shown in examples of RE planning in Norway and Denmark. Similarly, in a comparative study of Swedish municipalities' role in advancing energy democracy, Palm et al. [24] observe tensions between different normative goals such as direct participation of citizens versus more representative and expert guided understandings.

Another type of normative tension relates to conflicting understandings of nature and sustainability, often referred to as “green-on-green” conflicts [25,26]. For instance, climate mitigation efforts may be perceived as conflicting with local biodiversity or nature conservation efforts [27]. These green-on-green conflicts often do not simply replace traditional “black-on-green” conflicts (e.g., economic vs. environmental interests) but rather add new layers of complexity. While recent research has begun to address these tensions, further exploration is needed to understand how different dimensions of sustainability interact and occasionally conflict in RE planning.

Emphasising democracy in the RE transition resonates with broader scholarship on public participation. Public participation can be defined as “the practice of involving members of the public in the agenda-setting, decision-making, and policy-forming activities of organizations/institutions responsible for policy development” [28,p. 53]. This concept covers a wide range of participatory activities and formats from feedback surveys to continuing dialogues [19]. However, as Rowe and Frewer [28] note, the public participation domain often lacks clear definitions of key concepts and mechanisms for enacting participation. Various normative typologies exist, often inspired by Arnstein's seminal ‘ladder of participation’ [29] or by more recent frameworks such as Carpentier [30], who emphasises the equalization of power relations as central to participation processes. Across these typologies, three main forms of public participation are commonly identified: information, consultation, and cooperation [20,28,31,32]. Information represents the lowest level, involving only a one-way flow of information from the organising body to the public. Consultation occupies a middle ground, where communication is used to gather citizens' opinions on specific issues. Cooperation reflects the highest level of participation, enabling joint decision-making and mutual influence.

While mainstream research on participation has concentrated on single events or practices, such as town hall meetings or deliberative formats, research has begun to widen the perspective to broader systems of participation. In a pioneering work, inspired by research within Science and Technology Studies, practice theory and deliberative democracy, Chilvers et al. [33] develop the notion of ‘ecologies of participation’ to better account for the complexities and diversities of participatory energy practices. The authors apply their framework to a wide range of examples in the UK, ranging from emerging or decentred practices such as speculative design and sentiment mapping to dominant practices such as public consultation and public opinion surveys. Moreover, they distinguish interrelations between participatory practices, for example replication, assistance, and transformation. These interrelations help stabilize, or challenge, the wider ecologies of participation and energy.

In the present study, we side with this ecological, i.e. relational and systemic, approach and with the critique of a perspective on participation as occurring in discrete events or processes and as engaging primarily (groups of) individuals [33], also in line with the systemic turn within deliberative democracy theory [34]. At the same time, we take a different direction and narrow down the focus to a specific site and the interconnections between diverse communicative events related to that site, in casu to the participatory initiative of the Thy Model. While Chilvers et al. [33] provide a broad mapping of participation practices across various (geographical) locations, the present study concentrates on one site and the interconnections between various communicative events related to that site. Thus, this study responds to the desideratum articulated in Chilver et al. [33,p. 209] by offering in-depth empirical insights into a situated ecology.

This further implies a focus on matters of communication and their role and implications in the context of public energy participation. Thus, the contribution of the present study lies in providing insights into communicative dynamics of participation that may be used to design, facilitate, and evaluate communicative measures. The relevance of a communicative focus is illustrated by the fact that a significant part of the public resistance issues are related to communication, i.e. lack of democratic involvement, lack of transparency in decision-making, and insufficient procedural processes At a more fundamental level, we view communication not at as a simple transfer of messages but as socially constitutive, both regarding how people make sense of the world and how they build social relations and identities.

2.1. The communication ecology framework

As indicated above, we investigate the communicative dynamics of the Thy model as a network of practices, in which activities occurring at different times and locations are interconnected. Rather than focusing only on isolated communicative events such as a citizen meeting, we broaden the analytical lens to conceive of the Thy Model as an emergent communication ecology that situates and connects individual events within a wider dynamic [35]. The concept of communication ecology was originally developed within information technology and medium theory to "help grasp how social activities are joined with information technology and to offer a perspective for reconceptualizing how communication frameworks can inform social participation" [35,p. 666]. It has since been applied to diverse fields including community participation [36] and political communication [37]. Despite its growing use, the full potential of the concept remains underexplored. We argue that citizen participation in the RE transition provides a fertile ground for further conceptual enrichment from communication and discourse studies.

A communication ecology can be defined as "a milieu of agents who are connected in various ways by various exchanges of mediated and unmediated forms of communication" ([38] quoting [39]). We employ communication ecology as a sensitizing concept that through its biological metaphor highlights relationality, emergence, process dynamics, and temporal and spatial interconnections [35]. Applied to the Thy case, this perspective allows us to identify and analyse diverse forms of distributed participation across multiple sites and interactions. The communication ecology framework distinguishes three interrelated layers within a communication ecology: social, technological, and discursive.

First, the social layer encompasses "the social modes of organizing […] people", including "everything from friendship groups to more formal community organizations" [38,p. 1]. This layer addresses the relational and organisational dimensions of the ecology, incorporating the social actors and the connections between communicative activities distributed in time and space. Second, the technological layer comprises communication technologies and, in our interpretation, mediational means [40] that enable, constrain, or afford particular modes of interaction. This includes specific communication formats or interaction orders in Goffman's terms, such as a "with," a "meeting," or a "platform event" [41], which provide structure and meaning to the interaction. Third, the discursive layer pertains to the content of communication, namely, the ideas, narratives, and meaning-making processes involved. Drawing on Hajer [42], we understand this layer as concerned with how "meaning is ascribed to social and physical phenomena", for instance by "notions, ideas, concepts, and categorizations" [42,p. 60]. In that sense, the discursive layer includes the forms of knowledge that emerge in communication, as well as the identities and forms of agency enacted by participants. In line with the sociology of knowledge approach to discourse [43], we treat knowledge not as a fixed entity, but as discourse that is recognized and validated as knowledge by participants.

3. The role of municipalities in Denmark's climate politics

RE has been high on the Danish political agenda for many years, driven by concerns about global warming and energy security [44].

Climate politics in Denmark is closely aligned with international and European efforts to reduce greenhouse gas emissions and mitigate climate change, notably the Paris Agreement (2015) and the European Green Deal (2019). In the development and utilization of RE sources, particularly wind power, Denmark holds a leading position globally. In 2025, Denmark generated most of its electricity from RE sources. Wind accounts for close to 59% of electricity generation in 2025 [45]. Solar power and bioenergy also contribute significantly with 13% and 14% respectively [45]. Overall, low-carbon sources make up 86% of Denmark's electricity consumption [45], corresponding to 46.5% of the total energy consumption in 2024 [46]. Wind, solar, and bioenergy are expected to drive the transition to clean energy and increased electrification.

In recent years, the Danish government and other state-level institutions have introduced several ambitious climate policies, agreements, laws and regulations. These include, for example, the Climate Act adopted in 2020, which sets targets of a 70% reduction in greenhouse gas by 2030 compared to 1990 levels and climate neutrality by 2050 [47]. In addition, a 2023 climate agreement aims to quadruple RE capacity on land by 2030. In 2024, a green tripartite agreement was concluded between the Danish government and interest groups representing agriculture, nature conservation, trade unions, farmer associations, industry, and municipalities. This agreement primarily focusses on improving water quality, reducing carbon emissions from agriculture, and allocating significantly more space for nature, including forests and natural habitats [48].

Achieving the established climate goals requires coordinated implementation efforts at both central, regional, and local levels. At the local level, municipalities play a crucial role in implementing the measures necessary to meet these goals. Each municipality is governed by an elected municipal council and supported by a municipal administration. In addition to delivering local public services and providing social benefits such as education and elderly care, municipalities are responsible for areas such as local roads and public transport, energy supply and utilities, and land use planning [49]. As land use planners, co-owners of local utility and transportation companies, and major energy consumers, municipalities have significant influence on developments in these areas. Consequently, Danish municipalities play a key role in advancing the green transition, particularly due to their proximity to local citizens and communities. As part of their mandate, municipalities carry out strategic short- and long-term spatial planning that takes national interests into account. In addition to the municipal plan, most municipalities have also developed their own climate action plans to support national climate ambitions. This positions municipalities as powerful drivers of the green energy transition, although the task is complex and resource intensive. It is important to note, however, that differences between municipalities, such as whether they are primarily rural or urban, affect the opportunities and constraints for locating RE facilities. In addition, challenges such as population decline, job losses,

and decreasing tax revenues vary across municipalities. Finally, other pressing issues and geopolitical developments may threaten the prioritisation of the green transition.

4. Research design, case and data

To provide a foundation for the analysis, this chapter offers a case description, presents the research design and empirical data, and explains the analytical approach. The study employs a case study research design [50,51] with a qualitative, multi-source approach to exploring potentials and limitations for citizen participation within the Thy Model's communication ecology.

The selected case, Thisted Municipality, including the project site Bjerre, is located in northwestern Denmark along a 226 km coastline. The municipality covers approximately 1000 km^2 and has a population of about 43,000 ($\approx$ 40 inhabitants/km^2 [52]. The municipality faces demographic challenges, including a declining share of young people [53]. It is predominantly rural, characterized by unique landscapes, parts of which are protected within a national park. Key economic sectors include fisheries, agriculture, food processing, metal and plastics industries, and tourism [54]. Energy supply is largely based on renewable sources: about 80% of electricity is generated from wind power, supplemented by biogas, solar, and geothermal energy. Historically, the transition towards 100% renewable electricity has been driven by grassroot initiatives and private investments since the early 1980s [55]. Currently, the green energy transition is driven primarily by municipal ambitions for achieving CO_2 neutrality by 2035, enhancing climate resilience, advancing electrification, promoting green growth, and attracting and retaining residents and businesses [56] alongside the national climate objectives outlined above.

Following years of substantial local resistance to proposed RE projects, the Municipality of Thisted significantly restructured its approach to the green energy transition emphasising dialogue and citizen participation [56]. Over the past decade, only eight wind turbines were installed in the municipality. The most recent planning cycle proved highly contentious: from around 35 project applications, only one advanced after a year-long process generating more than 1000 pages of consultation responses and numerous access-to-information requests. The conflict overwhelmed the municipal administration, and consequently the municipality halted all wind and solar development for several years, prompting a reassessment of its approach to governance and citizen participation. As a response, the adoption of the Thy Model by the municipal council in 2023 seeks to leverage the development of RE facilities as a driver of local development and long-term community value, while explicitly encouraging local initiatives and civil society mobilisation [56]. This makes the Thy Model an interesting and suitable case for examining more participatory ways of engaging local communities and citizens in RE development.

To examine the social, technological, and discursive layers of the Thy Model and enable triangulation of findings, we have engaged with the field in various ways and integrated diverse data sources including semi-structured interviews, observations, policy documents, and media articles. Data collection took place between March 2024 and June 2025, both onsite and remotely. Appendix 1 provides an overview of the primary empirical material, specifying interviews, observation and document sources drawn upon in the analysis. Supplementary sources such as additional interviews, policy documents, media coverage, and social media communication related to RE planning in Denmark were used as background information but are not explicitly referenced in the analysis. To aid recall and ensure accuracy, all interviews (except INT1 and INT2) were audio-recorded, transcribed, and complemented by notes. Observations at citizen meetings were systematically documented in detailed observational notes.

The following analysis is structured by the three communicative layers of the communication ecology framework, seeking to uncover how patterns of meaning are constructed, and how participatory processes are shaped through these dynamics. We apply the communication ecology framework at two levels: the municipality as a whole and a project site. First, we analyse the Thy Model at the municipal level, focusing on how key organisational elements of the new communication ecology are established. Second, we turn to the Bjerre site, analysing the debate and dynamics surrounding a proposed PV plant as a local manifestation of the Thy Model.

5. Analysis

5.1. The municipal level. The social layer

The adoption of the Thy Model by the municipal council in 2023 marked a fundamental shift in the roles and responsibilities of both civil society, project developers, and the municipal administration. At the social layer, this implied that new organisational entities were introduced. A first step was the creation of Team Energy and Resources (TER) in early 2023, an interdisciplinary team of eight municipal employees, established to provide expertise, guidance, and facilitation of networking and capacity-building activities, particularly in relation to the local citizens' and communities' negotiations with project developers. TER was exempted from regulatory processing tasks to enable a more flexible, facilitative role (INT2).

As the next step, TER played a central role in the formation of eight so-called 'energy associations' at the local level by approaching already existing associations such as community associations and village guilds as well as engaged citizens with relevant competences (INT2, INT3, INT4, INT5, DOC2, DOC5). The outreach included both formal and informal meetings, as described in Section 5.1.1, as part of a long-term trust building process (INT2). According to TER, their own role was to facilitate the organisation of the energy associations, while refraining from influencing the content of discussions and negotiations with developers (INT2). The energy associations, established between June 2023 and June 2024, were intended to organise and represent citizens in the local communities on matters related to RE, to serve as an arena for dialogue, and to represent the local community in communication with both the municipality and the energy project developers (INT2, INT3, INT4, INT5, DOC2, DOC5). Finally, external energy consultants were hired as impartial advisors for the energy associations on legal and economic matters, such as compensation (INT2, OBS3). With the aim of strengthening the community's position vis-à-vis project developers, these actors should act independently of the municipal council and administration.

The establishment of these three social actors and their respective roles have significant implications for the organisation of communication between citizens and project developers. It is directed towards strengthening capacities for local communities and facilitating the dialogue with project developers, rather than directly organising specific meetings between local citizens and developers. Thus, the social layer at the municipal level represents a way of influencing local RE negotiations from a distance (Fig. 1).

5.1.1. The municipal level. The technological layer

The development phase of the Thy Model was characterized by several parallel communicative processes and formats. One of TER's initial tasks was to involve and gain input from citizens and communities, not about specific RE projects, but about how citizen involvement in the green energy transition should be organised across the municipality, and what principles should guide participation and decision-making. In September 2023, two large public meetings were held across the municipality to engage in dialogue with citizens and explore opportunities and barriers, with a strong emphasis on how to generate local value from large-scale projects. According to the TER, these meetings elicited reactions such as 'Why are you asking us now, unlike with the school closure decisions?' and 'Why do you ask how we want to be involved? Can we not just discuss solutions?' (INT2). This highlighted

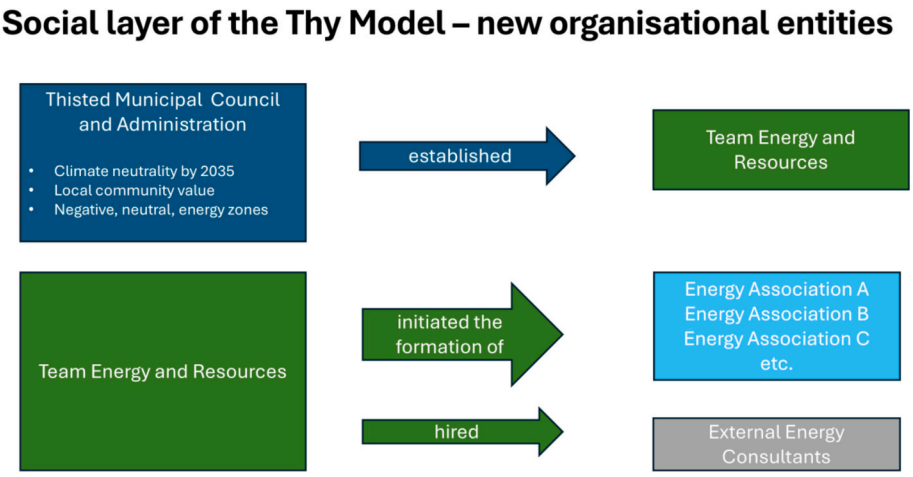


Fig. 1. The social layer of the Thy Model.

a need to build trust in the municipality. Around the time of the public meetings, a survey was sent to all 32,000 residents aged 18 and up, which received over 4000 responses (DOC4). Third, the TER reached out to around 60 local civic associations, visiting many during their regular meetings or informal community dinings (INT2). In addition, pilot projects were launched, enabling citizen-driven identification of suitable areas for RE plants. The interviews with TER employees (INT1, INT2) indicate that the approach was motivated by a desire for open processes, including access to internal municipal documents and a more equitable dialogue on finding location for RE plants.

Overall, the technological layer of the Thy Model's communication ecology reflects a long-term approach using diverse communication formats (Fig. 2), addressing individuals as well as community groups, and combining immediate feedback with more reflective responses. It can be characterized as a pre- or meta-involvement since it organises early-stage involvement of different citizen publics about how to organise local participation at a later stage.

5.1.2. The municipal level. The discursive layer

At the discursive layer, we focus on the main rationales and legitimations in the municipal presentation of the Thy Model. The formation of energy associations is mainly articulated within a discourse of local democracy and citizen involvement, using verbs such as "enter into a dialogue", "listen to" and "involve" (DOC1). The energy association is also referred to as a "common voice" for the local area that "strengthens the position of the local community" (DOC1). In interviews with TER, this is further backed with reference to the otherwise often unequal power balance between an energy developer with strong technical, judicial and economic resources and, on the other side, a local community with little experience with energy plants. Thus, the formation of energy associations is viewed as a way of restoring the basis for a "more equitable dialogue" and to "equip the local community to engage in this dialogue with the developers" (INT 2).

Another key point in which rationales and legitimations become visible, is the formulation of five guiding "themes" for the political assessment of RE proposals. Based on the resident survey and the interactions with citizens and local associations, TER developed five "political themes" (DOC1) that serve as criteria for assessing RE projects (see Fig. 3). These were later endorsed by the municipal council. The aim was to guide project developers and help the municipal council prioritise projects that align with citizens' values. These criteria reflect both economic, environmental, and democratic concerns: Theme 1 centres on economic ownership and demands that a minimum of 50% of the RE facilities are owned by the local community. In a less specific manner, theme 2 calls for fair compensation according to the principle that those who are most affected should receive the greatest compensation. Theme 3 requires contribution to nature improvement, such as afforestation or green corridors for wild animals, and theme 4 calls for adjustment to existing landscape qualities. Finally, theme 5 states that broad local support is key, and that local ballots will be arranged by the municipality. In addition, other indications of local support such as written commitments from neighbours or votes within energy associations are invited from the project developers.

The five criteria have a double discursive status: First, they are of course discursive representations of the Thy Model, combining economic, environmental, and democratic rationales. Second, they also set

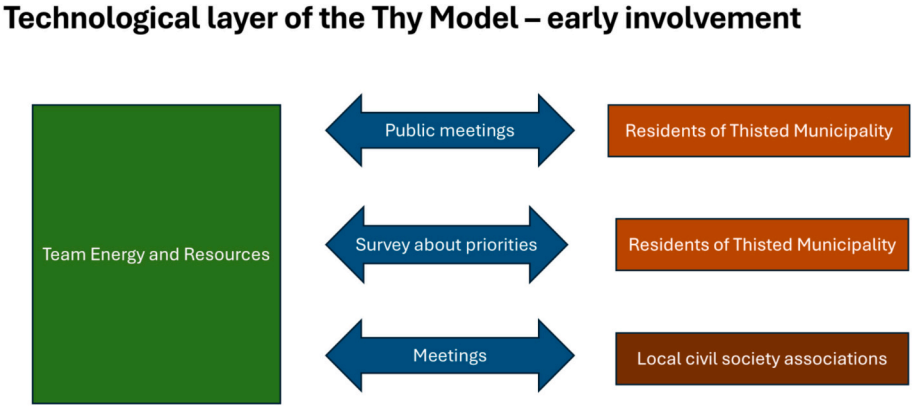


Fig. 2. The technological layer of the Thy Model.

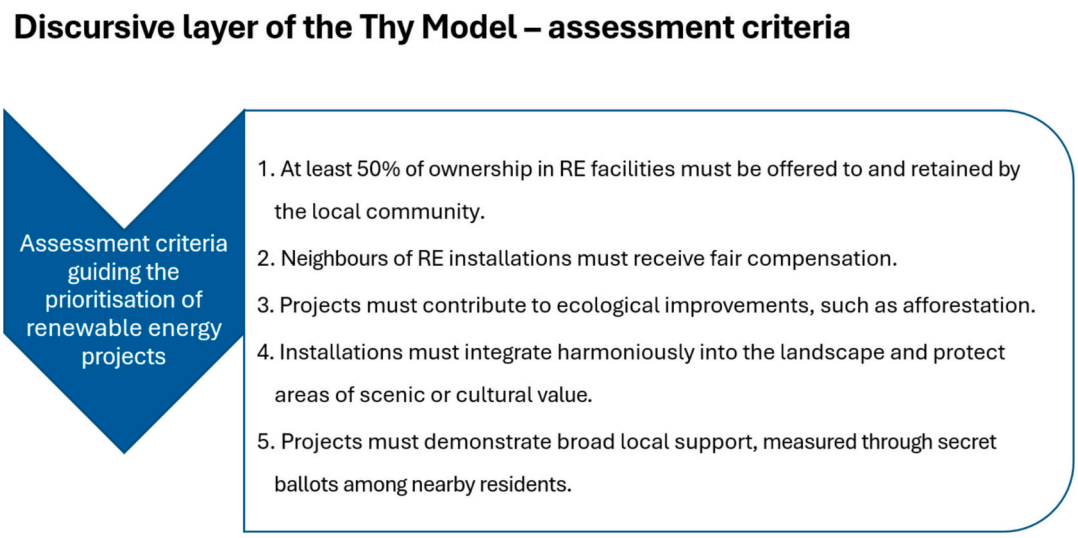


Fig. 3. The assessment criteria of the Thy Model.

conditions for the interactions within the communication ecology. While the criteria do not directly prescribe a certain outcome for the negotiations between citizens and developers, they indirectly regulate these by construing a discursive space that sets the conditions of acceptability.

5.2. *Project site: Bjerre*

Since the launch of the Thy Model, 13 RE project proposals have been developed by a range of actors, including private landowners, civic associations, and energy companies. These proposals were presented at citizen meetings (e.g., OBS1, OBS4, OBS6) and published on the municipal website. In March 2025, eight of the 13 projects were put to local ballots, with voter turnout ranging from 22.4% to 60.8% (DOC1). The results were decisive in every case: four projects received clear majority support, while the other four were clearly rejected. The ballot results, including location, type of RE, number of eligible voters, and ballot results, are shown in Appendix 2. In April 2025, the municipal council decided to advance the four approved projects to the next stage of the planning process, including the Bjerre project (DOC1, INT10).

Bjerre is located in the northwestern part of Thisted municipality, near Thisted Airport, the harbour town of Hanstholm, and Thy National Park. The project area includes 324 eligible voters (OBS2). The proposal involves expanding an existing solar power plant adjacent to Thisted Airport. It is developed by European Energy, a global energy company founded in Denmark in 2004, in partnership with Thy-Mors Energi, the local utility cooperative. According to the developers, the project will cover approximately 237 ha, including 25 ha for afforestation and 12 ha for green areas and wildlife corridors. The 32 nearest neighbours have been offered compensation for potential property value loss. Additionally, up to DKK 20 million will be allocated to a green fund, of which 80% will benefit the local community. This is supplemented by an annual contribution of DKK 333,000 over the facility's expected 30-year lifespan. Local ownership may be secured through a 25% purchase option in the project (DOC6, DOC8).

Bjerre has been selected for analysis because it represents a relatively average case among the eight election sites in terms of RE type, number of eligible voters, and voter turnout. Moreover, the strong vote in favour of the project proposal makes Bjerre well suited for examining the participatory potential of the Thy Model, as it suggests that local engagement dynamics were sufficiently active to reveal how the model functions in practice.

5.2.1. *The project site. The social layer*

The social layer provides an overview of actors and connections within the local communication ecology, using the two citizen meetings as reference points. These meetings stand out by involving most key actors: the TER, the developers European Energy and Thy-Mors Energi, the local Energy Association, an external energy consultant, and numerous local citizens. Invitations (DOC7, DOC9) were sent digitally to all eligible voters, and the meetings were hosted by the developers.

At the first citizen meeting in November 2024 (OBS2), a TER representative presented the Thy Model and situated the meeting within a broader community participation process, which included the following elements:

- The local energy association was introduced as both a facilitator of dialogue within the community and a spokes group representing local interests to the developers and municipality.
- A timeline for the participation process was outlined, comprising a three-month dialogue phase followed by an eight-week official hearing phase, concluding with a local ballot on the project (DOC8).
- The criteria for the municipal council's decision on whether to advance the project to the planning phase were presented, with local support, measured primarily through the ballot, emphasised as the most important.

The TER representative thereby outlined a communication ecology with the newly established local energy association in a key position between local citizens on the one hand and developers or municipality on the other. Moreover, in the outlined ecology, citizen participation is extended beyond the two citizen meetings to encompass a broader range of interactions, distributed over approximately five months, and involving the energy association, the developers, and local citizens. However, apart from references to the official hearing process and the local ballot, these interactions were not explicitly referred to.

Following the TER presentation, a developer representative also presented (DOC8) at the first citizen meeting, calling for dialogue with the local community in the coming months. In the same vein, the developer representative responded to questions raised in earlier communications with citizens and positioned the developer as a company that had “started afresh”, now enabling citizens to influence projects at a much earlier stage, pointing out: “It's up to you whether it will happen, there will be a ballot” (OBS2). The participating citizens mainly commented on and asked questions about the project's local impact and content (cf. Section 5.2.3) while some, such as a local business representative, expressed dissatisfaction at not having been involved earlier.

The second citizen meeting (OBS5), held nearly three months later, offered insights into how the communication ecology had evolved. In addition to presentations on the Thy Model procedure and the specific project (DOC1), similar to the first meeting, several comments referred back to interactions during the intervening months. First, the developer representative mentioned communications with various actors including neighbours to the planned PV plant, local businesses, the Danish Society for Nature Conservation, the local Motocross Club, and a nearby small airport. Second, an external energy consultant, who moderated the discussion, referred to extensive negotiations between the energy association and the developers. Third, a representative from the partner developer Thy-Mors Energi described a walk organised a week earlier, where approximately 25 citizens participated and suggested changes to the project's physical design (DOC9). Building on this, a landscape architect who joined the walk presented drafts incorporating citizens' ideas. Table 1 provides an overview of connections between the two citizen meetings and past and future communicative events referred to at the meetings.

The dense connections to other communicative events indicate that citizen participation extended well beyond the two citizen meetings. Instead, participation appears to have been distributed within a broader communication ecology with developers and the energy association as key actors engaging with local citizens, organisations, and businesses, and not least, with each other. Interviews with citizens and one developer support this view (INT7, INT9). Additionally, another local group was formed alongside the energy association: approximately 30 neighbours living near the proposed PV plant, 'local-locals' so to speak, organised independently (INT7, INT9). To secure sufficient representation of the interests of this particularly affected group, a few citizens entered into a series of direct negotiations with the developer regarding compensation (INT7).

5.2.2. The project site. The technological layer

Both citizen meetings were held in a sports hall accommodating just over 100 people, located in a former school building near the proposed project site. Each event followed a similar format: presentations using PowerPoint slides (DOC8, DOC11) were delivered by speakers and were followed, or occasionally interrupted, by questions and comments from attendees.

Table 1
Overview of interdependencies among communicative events at the project site.

Citizen meeting pointing back to:		Citizen meeting pointing forward to:
• Survey • Municipal planning of the Thy Model • Municipal decision on 80% of 'Green Fund' to local community • Municipal team's contact to civil society organisations • Formation of energy associations • Communication between project developers and some local citizens • Local landowners reaching out to project developers	First citizen meeting about the Bjerre project 26.11.2024	• Dialogue on the project between energy association, local citizens and developers • The hearing process • Habitat assessment by biologists from the developer • The ballot
• Energy association's negotiations with developers • Site walk with landscape architect, local citizens and developer • Developer's dialogue with stakeholders such as airport, motocross club, and The Danish Society for Nature Conservation • Dialogue between energy consultant and energy association • Local community dialogue (citizens–citizens)	Second citizen meeting about the Bjerre project 25.2.2025	• A 'local-local' group living close to the project site • A follow-up group to monitor the project development if realized • The ballot

At the first meeting in November (OBS2), chairs were arranged closely together in rows across the hall, all facing a presentation screen at one end of the room. Speakers took turns presenting slides and giving their talks. However, the screen was positioned too low for attendees in the back rows to see clearly, and some reported difficulty hearing. It was mentioned that the slides would be shared on the municipal project site afterward. With over 100 attendees, the hall was packed, and the atmosphere was dense and slightly tense, with a strong focus on the speakers. The room setup and communication format encouraged one-way communication more than dialogue, despite invitations for questions. The setting may have contributed to the fact that the speakers came under pressure, and that the tone was rather confrontational. Although a break with coffee, soft drinks, or beer had been announced, the presentations ran over time, partly due to audience questions and comments.

At the second citizen meeting in February (OBS5), the seating arrangement was notably different. Chairs were placed in half circles facing the screen at the long end of the sport hall, positioned closer to the screen and spaced further apart. This setup made it easier for attendees to see and hear one another. Around 35 people attended. Compared to the first meeting, the arrangement created a greater sense of space and ease. The atmosphere was lighter and more relaxed, with more dialogue among the attendees, not just between them and the speakers.

In addition, the invitations to the citizen meetings contributed to shaping the dynamics the meetings. Notably, the first meeting was framed as an 'information meeting', whereas the second was described as a 'citizen meeting'. From a participation perspective, these framings may have influenced the expectations, behaviours, and roles of both participants and organisers, as an 'information meeting' implies a more passive participant role with less engagement than a 'citizen meeting'.

Moreover, the way in which the invitations were distributed appeared to have caused some confusion among the invitees. Both invitations were sent via the official digital mailbox used by public authorities. Formally, the invitations originated from the project developers, reflecting the Thy Model's allocation of responsibility to developers for demonstrating local support. However, because the invitations were distributed through the municipality, it remained unclear to some participants who was inviting them and for what purpose. This ambiguity became evident at the first meeting, where a participant pointed out that the invitation lacked clarity regarding the intended recipients, the exact location of the meeting, and relevant contact details.

5.2.3. The project site. The discursive layer

A plethora of issues were brought up during the two citizen meetings. Our analysis of the discursive layer centres on three main themes: the impact on the local nature, the organisation and implications of the ballot, and economic compensation for neighbours and the community. These themes reflect the participants' expressed priorities and span environmental, economic, and democratic discourses.

At the first meeting (OBS2), the developers introduced the project as a potential boost to biodiversity, citing plans for afforestation and wildlife corridors linking two large natural areas, as well as new recreational opportunities for local communities. Citizens, however, voiced strong criticism. One argued corridors were too narrow, based on his observations of local deer movements. Another questioned the quality of the proposed nature, whether it really was going to be "great nature", while a third noted that the RE project would be situated "very close" to a natural park. Yet another pointed to an existing PV plant in the area and stated that animal life there is highly restricted. These critiques reflect not only concerns for nature and biodiversity but also draw on participants' situated experiences and local knowledge.

The second citizen meeting (OBS5) revealed notable discursive shifts regarding the impact on local nature. First, a landscape architect hired by the developers presented different landscape scenarios, including ideas from the prior walk, such as maps of recreational paths and safer traffic routes between villages. Second, citizens discussed which tree species would be suitable for planting around the PV facility. Third, one citizen suggested measures to allow deer movement around the PV facility. Thus, while the themes remained, we can observe a bidirectional shift: developers adapting to local input and citizens engaging in co-design. In discourse terms, the earlier strict opposition between green energy transition and concerns for biodiversity and nature gave way, at least partly, to negotiations over their compatibility.

Regarding the ballot issue, the TER representative presented its rationale at the first meeting, drawing on a discourse of democracy and community participation. According to this perspective, the ballot serves to express and measure the community's level of support, or lack thereof, after a period of information and debate. However, this was challenged by several citizens, who raised questions about the ballot's status within the municipal council. One citizen asked whether a majority vote against the projects would automatically compel the council to reject them. The TER representative declined this, explaining that the interpretation of the ballot results ultimately constitutes a political assessment, although she added that she could not imagine the council disregarding a clear majority. This answer triggered visible frustration among several attendees and prompted a citizen to denounce the process as neither fully democratic nor transparent: “Then it doesn't matter; we can't have a democratic process when we don't know the percentages, if the politicians can simply interpret or decide as they please” (OBS2). Similar criticism of the ballot as a tool for democratic participation is can be observed in both an interview with a local citizen (INT8) and in media articles (e.g., DOC10, DOC13). Conversely, the ballot results have also been interpreted as a way to give voice to otherwise silent citizens (e.g., DOC12, DOC14).

At the second meeting, the ballot issue was revisited, and an additional concern was raised. A citizen noted that both residents living adjacent to the proposed project and up to 1200 m away are eligible to vote and asked whether there was a way to give greater weight to the votes of those most affected. The TER representative clarified that this would not be possible, emphasising that the broader community would also be impacted by the PV plant. Consequently, no conciliatory discursive shift occurred regarding the ballot. Instead, democratic dilemmas surrounding the local voting process were explored, and diverse arguments were articulated and made visible throughout the discussion.

Finally, issues of economic compensation arose at both citizen meetings. The developer and TER informed about compensation from the Green Fund, noting that 80% of the amount would go to the community (OBS2, DOC8). The developer also outlined ownership structures and options for private investment, as well as voluntary agreement on value loss for the nearest neighbours. However, unlike other economic matters, the agreement's content was not disclosed. While information on the Green Fund and investment options drew no critical questions, some citizens challenged the legitimacy and fairness of the individual compensation. At the same time, indications of improved compensation terms emerged at the second meeting. The external energy consultant thanked the energy association for negotiating better deals for residents not living immediately adjacent to the proposed project. According to one local citizen (INT7), the representatives of the ‘local-locals’ had also secured significantly improved compensation directly with the developers, bypassing the energy association. Although details were not made public, compensation appears to have been a key issue and one that progressed through interactions between developers, the energy association, and the ‘local-locals’.

6. Conclusion and discussion

We start by concluding on our first research question concerning how the social, technological, and discursive dimensions of the Thy Model's communication ecology are shaped.

At the municipal level, the social layer shows the introduction of a new set of social actors: the municipal TER, the eight local energy associations, and the hired external energy consultants. These actors engage in different communicative formats, i.e. the technological layer, including large public meetings, a survey, and meetings between TER and local associations. Regarding the discursive layer, TER and Thisted Municipality draw heavily on a discourse of democracy and local participation. Moreover, local community benefits and considerations for environmental impact play an important role as does the idea of paving the way for a more equitable dialogue between developer and local community.

At the project site level, we find a rich configuration of actors at the social layer. Central is the local energy association, which in the Bjerre case interacts closely with the developers. Moreover, developers and the energy association interact with different civil society groups, businesses, and individual citizens. The role of TER is more backgrounded and concentrated in the two citizen meetings, while the energy consultant advises the local community and the energy association ahead of negotiations with developers and takes on a moderating role in one of the citizen meetings. Finally, the energy association is supplemented by a group of ‘local-locals’ that in a self-appointed way represents the citizens living closest to the planned RE facility in negotiations with the developers.

Regarding the technological layer, the communication spans multiple sites and formats, formal and staged, such as the citizen meetings, and informal, such as conversations between energy association representatives and other citizens. It includes both spoken and written modalities, as well as unconventional activities like the organised walk and talk with a landscape architect. Thus, based on the social and technological layers, we can conclude that the local negotiations at the Bjerre site are distributed across a wide range of communicative settings. These activities involve several actors in intersecting connections, pointing towards the ballot.

Concerning the discursive layer, discourses similar to those used by TER in presenting the Thy Model can be observed at the project site, although in different ways. Within economic and environmental discourses, we can observe that issues are problematized by citizens, but also that discursive shifts occur with citizens in changed positions, such as contributing ideas to the landscape design around the projected RE facilities. However, in relation to the discourse of local democracy, citizens instead contribute by pointing to democratic limitations and dilemmas of the participation process, not least regarding the political weight of the ballot and the delineation of those eligible for voting.

Based on these findings, we argue that citizen participation is distributed across two overarching communicative spaces within the communication ecology: a space of negotiation and a space of capacity building (see Fig. 4). In the negotiation space, project site negotiations occur with developers and energy associations in central roles, as outlined above. In the capacity building space, the energy associations receive support from energy consultants through face-to-face counselling. More indirectly, TER's organisational setup and a network of energy associations in Thisted Municipality also contribute to the local capacity building.

Regarding our second research question on the implications of the emerging communication ecology for citizen participation, we conclude by outlining both potentials and limitations in relation to three key features of the Thy Model: energy associations, the ballot, and capacity building.

First, the local energy associations offer significant participatory potential by providing an organised, collective voice for the community with an influence on the RE projects through negotiations with the RE developers. Within existing typologies of participation [28,31,32], this can be understood as an instance of cooperation.

From a broader perspective of landscape or energy democracy, the

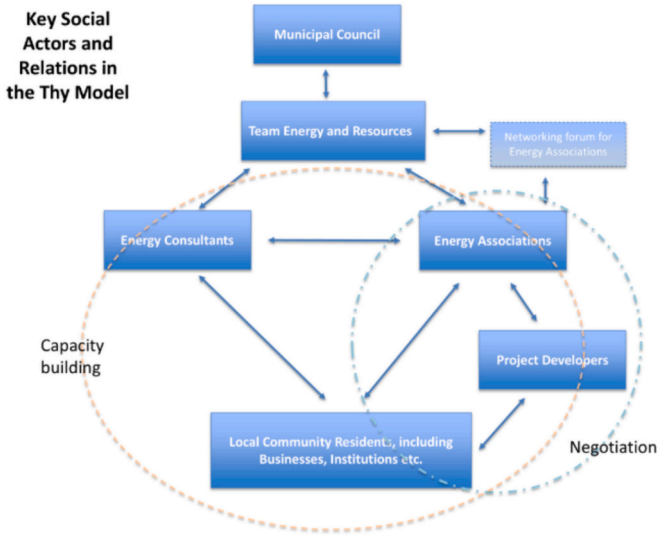


Fig. 4. Key social actors, relations, and communicative spaces in the Thy Model.

energy association's pivotal role aligns with what Arler et al. [23] describe as "co-determination and participatory rights" in their triangle model of landscape democracy. In contrast, the other two dimensions of landscape democracy, "personal freedom and private self-determination" and "objectivity and respect for arguments" [23,p. 7], are less pronounced in the Thy Model. This emphasis may reflect a response to RE projects that have failed due to insufficient citizen and community participation [16,23].

From an energy justice perspective [16,17], the central role of the energy association introduces a new institutional design with significant potential for enhanced procedural justice. The design relocates the weight of negotiations from citizen meetings or other forum events to a wide range of local interactions between energy associations, developers and other local stakeholders. Recognitional justice is also reinforced by the energy association's independence, which can help local perspectives, concerns, and knowledge to be voiced. This includes articulating diverse community values - economic, ecological, and social - concerning wildlife, landscape, traffic regulation, and civic life etc. While distributive justice is not guaranteed in the Thy Model, the Bjerre case indicates that some of these values can be integrated into the RE design, counterbalancing the plant's negative impacts.

However, questions can also be raised as to the legitimacy of the energy association representatives, specifically how they are selected and held accountable. In the Thy Model, most energy associations were established after TER approached individuals deemed capable, who then invited broader local participation. While this may provide a useful starting point, it calls for a stronger democratic foundation in the future. In fact, in the Bjerre case, the position of the energy association as representing the local community was challenged by the group of 'local-locals'. This group emerged to correct what was viewed as an insufficient representation of the nearest neighbours to the PV facility. This shows that the energy association is no institutional guarantee of democratic community representation and recognitional justice. It illustrates that the existence of a community is not self-evident, but that different constructions and ways of "spatializing" communities [57] may co-exist and give rise to competition and struggle. Moreover, on a resource level, the work of representatives is voluntary, relying on the time and energy of engaged local citizens. While this reflects a strength of civil society, it also limits the scope of what these associations can be expected to achieve.

Second, the secret local ballot on a proposed RE project constitutes at least an instance of consultation [32] as it provides feedback from the local community to the municipal council. It also indirectly incentivises developers to propose projects that benefit the broader community, rather than merely a few landowners. Moreover, the ballot's participatory value is strengthened by its timing in the communication ecology, taking place after negotiations between energy associations and the developers, thus reflecting the outcome of a cooperative process. This is in line with deliberative understandings of democracy and with a system perspective within deliberative democracy theory [34]. Participation through the ballot is limited, however, as it serves only an advisory role for the municipal council. Measured against Arnstein's ladder of participation [29], the highest rungs, 'delegated power' or 'citizen control', do not apply to the Thy Model, since political power is not delegated to the local community. As the analysis shows, this merely advisory role of the ballot was met with strong criticism from several participating citizens in the Bjerre case. However, while this can indeed be considered restrictive at the village or district level, it can also be viewed as legitimate within the framework of municipal representative democracy. The status of the ballot thus illustrates tensions between different conceptions and scales of democracy: representative versus direct, and village versus municipal.

These democratic tensions also surface as voting eligibility becomes a contentious issue. While Thisted Municipality has chosen to include citizens living relatively far from the projected RE plants, some citizens argue that this procedure risks overruling those most affected nearby. This reflects complex trade-offs between inclusion and exclusion; broad inclusion may dilute the voice of those most impacted. Again, this illustrates the importance of not taking community delineation and representation for granted [17,57]. In this respect, the Thy Model's contribution may not be to resolve these issues but to make them visible for broader discussion and further development and research.

Third, a key participatory potential of the Thy model's communication ecology lies in its emphasis on capacity building. Hiring an external energy consultant to advise the energy association on economic and legal matters is an effort to strengthen the energy association's ability to negotiate with developers on more equal terms. Also, the formation of energy associations itself fosters capacity building by enabling knowledge sharing and resource pooling within the community. Likewise, the municipal network of energy associations enables exchange across projects, thereby empowering individual energy associations. In this way, the Thy model deliberately counteracts existing power imbalances in line with suggestions from critical perspectives on participation [30]. However, these efforts do not eliminate disparities. For example, resources vary considerably, and new power imbalances may arise within the community.

Summing up, what is most relevant in the Thy Model from a communication and participation perspective may not be the individual elements of the model but their combination and interplay in a communication ecology. With some variations, similar elements can be found elsewhere in the field. For example, municipality organised ballots on local RE projects have been carried out recently in Denmark in the Municipality of Tønder, in the southwest of the country, and in Germany, citizen-initiated referenda have been used particularly in the areas of Schleswig-Holstein and Hessen [58]. Likewise, attempts to establish local groups like the energy associations to represent the local community in RE negotiations may not be exceptional. However, it is the combination of these elements that represents a novel approach of interest beyond the local case. To illustrate, in comparison with the mapping of participatory practices in the UK by Chilvers et al. [33], the Thy Model combines practices of public consultation, deliberative processes, community energy groups, and co-design. Moreover, these practices do not simply add up but also have a systemic impact. Thus, the ballot should be seen in relation to the negotiation work by the energy association leading up to the ballot. And the participatory value of the energy association should be seen in the context of the capacity building efforts carried out by the energy consultant.

We close with a brief reflection on limitations of the study and the

analytical framework. Regarding the latter, we see some overlap between the three layers that would benefit from further conceptual clarification and development in future research. Here, a stronger engagement with notions from discourse and communication studies, of which we have only pointed to a few [40–42], could be helpful. Moreover, the analytical split between the three layers ironically risks shifting attention away from the connections within the communication ecology. This risk therefore needs to be addressed with a synthesis approach alongside the analytical. Finally, the comprehensive ecology framework, in principle, calls for including all instances of communication related to the object of study. However, in practice it has not been possible to include, for example, everyday face-to-face and social media conversations, even though they may have contributed to the dynamics of the communication ecology as a whole. A further empirical limitation is that, due to time constraints, we did not include more project site cases and stakeholder perspectives, for example politicians and businesses.

CRediT authorship contribution statement

Anders Horsbøl: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anne Grethe J. Pedersen:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. Overview of data

ID	Type	Date	Social actor and context
INT1	Interview	22.3.2024	Two municipal planners, Thisted Municipality
INT2	Interview	23.4.2024	Two municipal planners, Thisted Municipality
INT3	Interview	4.7.2024	Head of Energy Association Nors, Tved, Vandet
INT4	Interview	8.8.2024	Head of Energy Association Brund-Skinnerup
INT5	Interview	12.9.2024	Head of Energy Association Sydthy
INT6	Interview	12.11.2024	One municipal planner, Thisted Municipality
INT7	Interview	5.2.2025	Resident, Bjerre, Thisted Municipality
INT8	Interview	6.3.2025	Resident, Bjerre, Thisted Municipality
INT9	Interview	13.3.2025	The head of Production, Thy-Mors Energi (project co-developer)
INT10	Interview	2.6.2025	One municipal planner, Thisted Municipality
OBS1	Observation	26.8.2024	Citizen meeting: Rosvang Energipark (Wind and PV)
OBS2	Observation	26.11.2024	Citizen meeting: Solcelleanlæg ved Bjerre (PV)
OBS3	Observation	27.1.2025	Online meeting: Experiences from VLF Rejseholdet (consultants)
OBS4	Observation	5.2.2025	Citizen meeting: Røjkær – Syd Thy Vind (Wind and PV)
OBS5	Observation	25.2.2025	Citizen meeting: Solcelleanlæg ved Bjerre (PV)
OBS6	Observation	25.2.2025	Citizen meeting: VE Tagmarken (PV)
DOC1	Municipal website	No date	Thisted Municipality's website: Thisted.dk (green transition, renewable energy, strategy, energy plan) https://www.thisted.dk/politik-demokrati-and-indflydelse/thy-til-groen-omstilling/vedvarende-energi#heading1
DOC2	Municipal article	29.11.2023	Thisted Municipality: 'Energiforeninger skyder frem i Thy' (Energy associations are emerging in Thy) https://www.thisted.dk/nyheder/2023/december/energiforening
DOC3	Minutes and reports	4.9.2023 11.3.2024 3.4.2024	Statements from municipal council meetings, Thisted Municipality
DOC4	Survey	Dec 2023	Thisted Municipality's analysis of questionnaire sent to 32,000 residents aged 18+
DOC5	Municipal article	March 2024	'Borgerinddragelse og gennemsigtighed kobler lokal udvikling og grøn omstilling i Thy' (Citizen involvement and transparency link development and green transistion in Thy) Teknik & Miljø, marts 2024. (link)
DOC6	Project application	30.5.2024	European Energy and Thy-Mors Energi: Project application about solar plant near Bjerre (dk.europeanenergy.com)
DOC7	Invitation	Nov 2024	Invitation to citizen meeting: PV plant Bjerre - Invitation til informationsmøde om solcelleanlæg ved Bjerre'
DOC8	PPT slides	26.11.2024	Presentations by municipal planners and project developers
DOC9	Invitation	Nov 2025	'Invitation til Gåtur og borgermøde om solcelleprojekt ved Bjerre' (Invitation to walk and citizen meeting on the Bjerre solar project) 21.2.2025 + 25.2.2025
DOC10	News media article	24.1.2025	'Kritik af simpel afstemning om store vindmøller' (Criticism of simple voting on large wind turbines). Nordjyske Stiftstidende.
DOC11	PPT slides	25.2.2025	Presentations by municipal planners, project developers, and landscape architect
DOC12	Press release	28.3.2025	Green Power Denmark: 'Det tavse positive flertal har fået en stemme i Thisted' (The silent positive majority has found a voice in Thisted)
DOC13	Opinion piece	7.9.2025	'Thy-modellen er bestået med et 7-tal' (The Thy-Model has passed with a grade of 7, i.e. good but not top). Nordjyske Stiftstidende.
DOC14	News media article	10.9.2025	'I Thy siger naboer ja til møller og solceller: Her er fem ting, der skal være på plads' (In Thy, neighbours say yes to wind turbines and solar panels: Here are five things that need to be in place). DR.dk.

Appendix 2. Ballot results for RE projects, Thisted Municipality, 2025

Location	RE source	Yes votes	No votes	Blank votes	Entitled voters	Turnout rate
Røjkær	PV + Wind	24,7%	73,2%	2,2%	2173	40,5%
Gisselbæk	Wind	60,8%	32,6%	6,6%	1352	19,1%
Bjerre	PV	63,9%	32,4%	3,7%	324	33,3%
Tagmarken	PV	34,5%	63,6%	1,8%	181	60,8%

(*continued on next page*)

(*continued*)

Location	RE source	Yes votes	No votes	Blank votes	Entitled voters	Turnout rate
Rosvang	PV + Wind	33,1%	63,6%	3,3%	1728	45,2%
Visby	PV	34,0%	62,4%	3,6%	2115	22,4%
Ballerum	PV	67,4%	18,6%	14,0%	79	54,4%
Vibbersdorf	Wind	59, 7%	35,7%	4,6%	1299	26,9%

Source: [59]

Data availability

Data is partly confidential, but some data can be made available on request.

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